Digital Closed-loop Fiber Optic Gyroscopes - A Choice for Micro/Nano-Satellites Application

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***Abstract*—Two improved compact architectures of three-axes IFOG using sharing components and time division multiplexing with a new modulation method were proposed. The miniaturization techniques for configurations, electronics and optical components were demonstrated. And an efficient approach of parameters monitoring and fault diagnosis has been developed. Finally the new products, performance test results and their uses in several micro/nano-satellites were introduced.**

Keywords—fiber optical gyroscope, digital closed-loop, miniaturization, micro/nano-satellites

I. Introduction

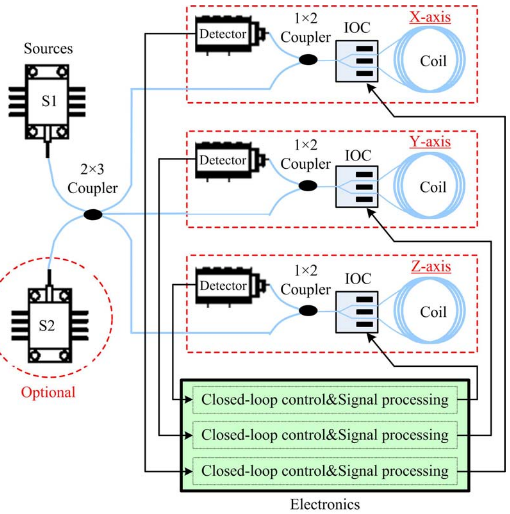
The interferometric fiber optical gyroscopes (IFOG) have already proved to adapt to space application because of their high performance, low weight, low power consumption and high reliability as a solid state technology, and widely served for various space missions in the past ten years [1–6]. The developed products and technologies were mainly oriented toward high precision and high reliability application in the long term space missions. As the use of small satellites is becoming increasingly common, the demand for miniaturized, low-cost and robust gyroscopes is surging. Mini IFOG is an available choice for small spacecraft including micro/nano-satellites, especially in harsh environments and high performance application.

Based on previous studies in our research group [4-12], the miniaturized spaceborne IFOG products have been improved continuously. This paper reports the recent progresses in research and development of the miniaturized digital closed-loop IFOGs and demonstrates two smaller products, ground and in-orbit tests results and their application in micro-satellites (10-100kg) and nano-satellites (<10kg).

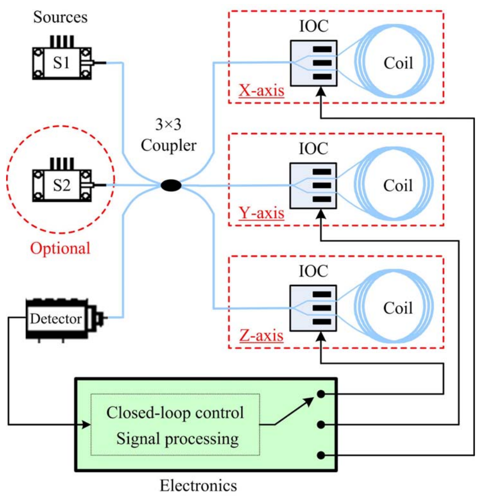
II. IFOG Configuration

For the purposes of high reliability and long life time, the four times or six times single-axis are the typical configurations in spaceborne IFOGs, in which the three mutually orthogonal axes to measure ration rates simultaneously are necessary for attitude control, navigation and orientation, and other independent axes are redundant [5]. But duplication of entire axes has resulted in significant increases of mass and power consumption, so this is not tolerable in many space missions like micro and nano-satellites.

Figure 1 (a) shows a typical configuration of three axes cluster. One light source and single digital signal processor are shared for all axes to cut the mass and power consumption, and the vulnerable source may be duplicated to guarantee the long term operations. That makes a compromise between reliability and miniaturization.



(a) Typical three axes cluster with source redundancy



(b) Time division multiplexing three axes

Fig. 1 Simplified digital closed-loop IFOG architectures

To further simplify the IFOG architecture, a minimum configuration of three-axis IFOG is illustrated in Figure 1 (b). A source is shared by three axes, and a photodetector, an 3*3 coupler and most electronics are multiplexed. Compared with the typical three axes cluster IFOG, this architecture removes two axes components except the fiber coils and integrated optics chips (IOC) by using time division multiplexing. It considerably decreases the mass, size, power consumption and cost of digital closed-loop IFOG. In this case, one more source is optional to enhance the vulnerable part.

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In these two IFOG configurations, polarization-maintaining (PM) fiber coils and super-luminescent diode (SLD) sources are adopted, the operational wavelength could be 1310nm or 850nm, and the IOC with proper modulation and demodulation methods achieves the all-digital loop closure operation and signal processing. The figure 2 gives a working sequence in time division multiplexing IFOG, the three axes are alternately working, and only one axis can be controlled to close loop and provide available rotation rate at one moment.

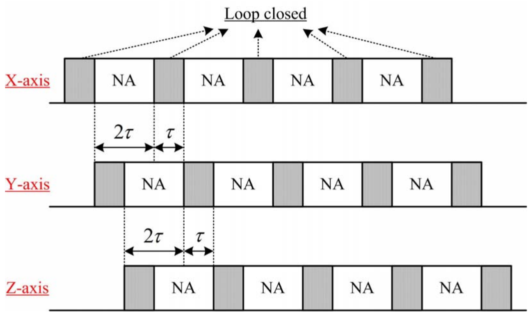


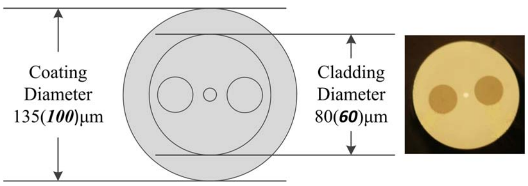
Fig. 2 Working sequence in time division multiplexing IFOG

III. IFOG MINIATURIZATION

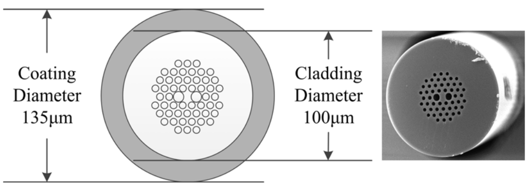
Besides simplifying the configuration, functions digitalization, circuits integration and optical components miniaturization are also important measures to shrink the IFOG's dimension. We have developed the miniaturized electronic devices and optical components such as integrated circuits, mini light source, small IOC and packaged PIN-FET. Compared to off-the-shelf components, their dimensions are decreased by half or more.

The dual closed-loop control circuits including analog-digital converter (ADC), digital-analog converter (DAC) and field programmable gate array (FPGA) have been integrated into one chip. A mini size circuit covering the photodetector, amplifiers and filters have been customized following our requirements to suit the IFOG miniaturization.

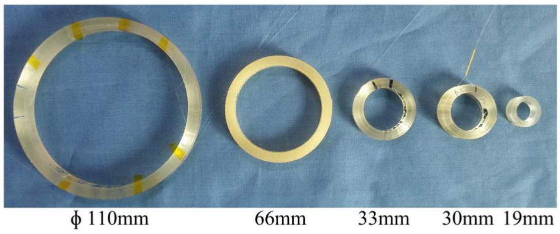
In addition, small diameter PM fibers and small diameter polarization maintaining photonic crystal fiber (PM-PCF) have been developed to decrease the size of fiber coils and improve the performance of miniaturized IFOG [7,8]. The fiber coating has reduced from 165μm to 135μm and 100μm, and the mini coils diameter is down to 19mm, as shown in figure 3.



(a) Small diameter PMF



(b) Small diameter PM-PCF



(c) Mini fiber coils

Fig. 3 Miniaturized optical components

Moreover, as status monitoring is indispensable in space missions, two efficient methods for vital parameters monitoring were proposed which removed the additional sensors and the associated circuits hardware and were implemented simply in FPGA. The new way measuring temperature in IFOG is based on the temperature dependence of the IOC, which is reliable, compactible and simple to be implemented [9]. Light power is also an important indication for changing source timely when the source power falls off. A novel method to monitor optical power was developed based on an algorithm of inverse-demodulation [10]. The above-mentioned two approaches are without hardware modification and only add a proportional integral derivative controller module, an inverse-demodulation module and a moving average module into the FPGA. These all digital implementations make IFOG's more compact. A parameters in-situ measurement in our space products is presented in figure 4.

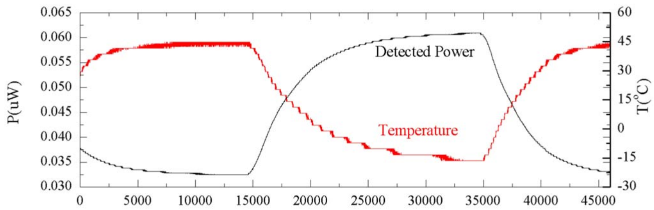
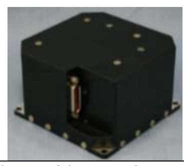
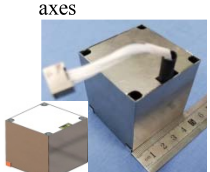


Fig. 4 In-situ measurement with all-digital monitoring methods

IV. PRODUCTS AND PERFORMANCE TESTS

Corresponding to above mentioned configurations, we have developed two space product models SF3X80-H and SF3X50, and their key specification and performance are listed in Table 1.

TABLE I. COMMERCIAL OFF-THE-SHELF PRODUCTS, PERFORMANCE AND APPLICATION

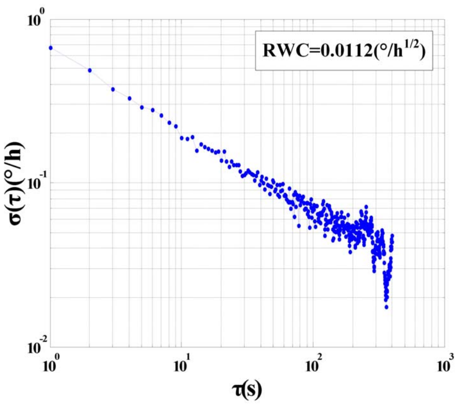
SF3X80-H	SF3X50
Three axes sharing one source 	Time division multiplexing three axes 
Size: 80mm×80mm×50mm Weight: ~0.43kg Power supply: +5V Power consumption: ≤4W Operating temparature: -40~60°C Interface: RS422 or CAN Output format: rotation rate, temperature, optical power	Size: 50mm×50mm×54mm Weight: ~0.2kg Power supply: +5V Power consumption: ≤2W Operating temparature: -40~60°C Interface: RS422 Output format: rotation rate, temperature, optical power

SF3X80-H	SF3X50
Bias instability: ≤0.05°/h RWC: ≤0.01°/√h Scale factor error: ≤100ppm	Bias instability: ≤0.3°/h RWC: ≤0.05°/√h Scale factor error: ≤200ppm
Commercial micro-satellite cluster ZHUHAI-1 (≤100kg) Scientific micro-satellite SPARK (≤50kg)	Nano-satellite ASRTU (≤10kg)

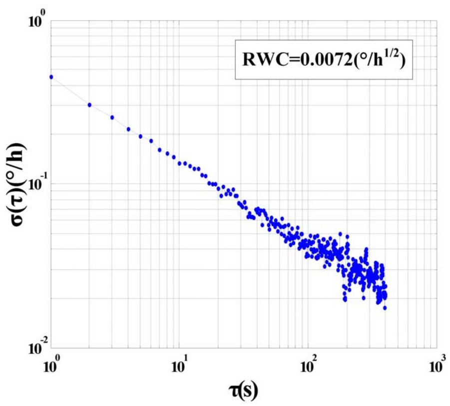
According to the requirements and constraints for gyroscopes in micro/nano-satellites, they have already passed the environmental tests and qualified for space application. The qualification tests briefly include:

- Shock and vibration (20Hz to 2000Hz)
- Thermal cycling (-30 ºC to +60 ºC)
- Thermal vacuum cycling (-30 ºC to +60 ºC)
- Storage at high and low temperature (-50ºC, +85ºC)
- Total ionizing dose (up to 20Krad)
- Magnetic sensitivity (≤0.02°/h/Gauss)

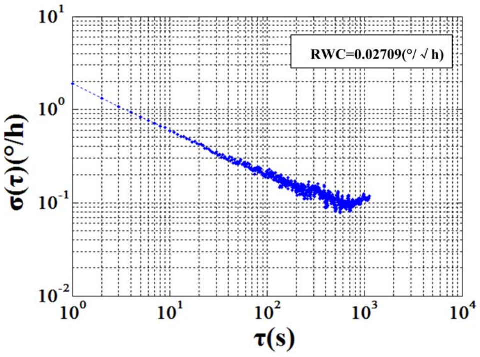
We conducted an in-orbit experiment to test the performance of model SF3X80-H. The in-orbit data of the IFOG were analyzed, and the random noise was extracted from the raw data [11,12]. From the refined data, the random walk coefficient (RWC) was calculated with the Allan variance method as shown in figure 4 (a). Allan variance ground test results for SF3X80-H and SF3X50 are demonstrated in figure 4 (b) (c).



(a) In-orbit test result for SF3X80-H



(b) Ground test result for SF3X80-H



(c) Ground test result for SF3X50

Figure 4. Allan variances of the IFOG products

SF3X80-H products from Beihang University have been successfully employed in commercial micro-satellite cluster ZHUHAI-1 (≤100kg) including 5 satellites for half year to stabilize the satellite platforms for imaging, and normally served in scientific micro-satellite SPARK (≤50kg) for almost 2 years. SF3X50 products have been delivered for Nano-satellite ASRTU (≤10kg), which will be launched in 2019.

V. Conclusion

Two compact three-axis configurations of digital closed-loop IFOG were proposed for micro and nano-satellites application. They are simplified by using components sharing and axes time division multiplexing. Miniature optical components, integrated circuits and all-digital implementation of health monitoring have been developed and applied to reduce IFOG dimension, cost and power consumption. Two very small commercial off-the-shelf products and their specification, performance and application are demonstrated. Both two have already been qualified for specific space missions, and the model SF3X80-H have been employed in orbit.

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